Observability of Private Energy Data in Coalition Formation via Inverse Optimization

Abstract

We investigate the observability of agents' private data—consumption/generation profiles and battery characteristics—in a coalition formation framework for buildings with battery storage. An adversary observing only the broadcast net grid trades (and, in coalition, the internal exchange quantities) can reconstruct the set of private parameters consistent with the observed behaviour. We formulate the inverse problem as a linear program derived from the KKT conditions of the forward optimization, and compute per-parameter observability bounds for both singleton and coalition settings. Our results show that (i) net load is partially observable from grid trades alone, with the battery creating significant ambiguity; (ii) coalition participation *increases* observability rather than decreasing it, because the coalition exchange vector reveals an additional equation per timestep; (iii) the existing differential privacy mechanism protects the coalition structure but not the private data revealed through the ADMM broadcast.

1 Introduction

Consider a set of N buildings, each equipped with a battery storage system. At each MPC step k, every building solves a receding-horizon optimization over h timesteps: minimise the cost of grid energy purchases (net of sales) subject to power balance, battery dynamics, and box constraints. The decision variables are the grid buy/sell quantities $g_{\text{buy}}[t], g_{\text{sell}}[t]$, the battery charge/discharge $\delta_s[t]$, and the state of charge $\text{SoC}[t]$.

Private parameters. Each building i has private data that it does not wish to reveal:

- *Net load profile*: $d_i[t] = \text{cons}_i[t] - \text{prod}_i[t]$, the difference between predicted consumption and generation at each timestep.
- *Battery capacity*: ${Q^b}_i$ (kWh).
- *Battery power rating*: ${P^b}_i$ (kW), the max charge/discharge rate.
- *Initial state of charge*: $\text{SoC}^0{}_i$ (kWh).

Public information. The following are known to all parties (including the adversary):

- Tariff: buy prices $p_{\text{buy}}[t]$ and sell prices $p_{\text{sell}}[t]$ for each timestep, with $p_{\text{buy}}[t] > p_{\text{sell}}[t]$.
- Battery efficiencies: $\eta_{\text{ch}} = \eta_{\text{dis}} = 0.95$ (uniform across all buildings; assumed known to the adversary).

Broadcast information. In the coalition formation framework (`privacy_focussed_coals`, `Coalition.jl:305`), each agent solves its MPC subproblem via ADMM and computes

$$\text{cons_vec}_i[t] = g_{\text{buy},i}[t] - g_{\text{sell},i}[t] \equiv z_i[t], \tag{1}$$

the net grid trade at each timestep. This vector is shared with the coalition coordinator to evaluate pairwise coalition benefits (`Coalition.jl:360,369`). In a coalition of two or more agents, the ADMM consensus variable—the *coalition exchange* $\xi_i[t]$—is also broadcast during iterations (`MPC_optimiser.jl:269`).

Research question. Given the observed quantities $z_i^*[t]$ (and optionally $\xi_i^*[t]$), what is the set of private parameters $(d_i, {Q^b}_i, {P^b}_i, {\mathrm{SoC}^0}_i)$ that could give rise to these observations? We call this the *observability set*, and we quantify observability via per-parameter bounds.

2 Inverse Optimization Formulation

2.1 Forward Problem

The singleton MPC subproblem at step k with horizon h is the following LP:

$$\min_{g^+,g^-,\delta^+,\delta^-,c} \quad \sum_{t=1}^{h} \big(p_{\text{buy}}[t]\, g^+[t] - p_{\text{sell}}[t]\, g^-[t]\big) \tag{2a}$$

$$\text{s.t.} \quad d[t] + g^-[t] + \delta^+[t] + \delta^-[t] = g^+[t] \qquad \forall t \tag{2b}$$

$$c[1] = \mathrm{SoC}^0 \tag{2c}$$

$$c[t] = c[t-1] + \eta_{\text{ch}}\, \delta^+[t-1] + \eta_{\text{ch}}^{-1}\, \delta^-[t-1] \qquad \forall t \geq 2 \tag{2d}$$

$$0 \leq \delta^+[t] \leq P^b, \quad -P^b \leq \delta^-[t] \leq 0 \qquad \forall t \tag{2e}$$

$$0 \leq c[t] \leq Q^b \qquad \forall t \tag{2f}$$

$$g^+[t] \geq 0, \quad g^-[t] \geq 0 \qquad \forall t \tag{2g}$$

$$\delta^+[h] + \delta^-[h] + c[h] \geq 0 \tag{2h}$$

where g^+, g^- are grid buy/sell, δ^+, δ^- are charge/discharge, and c is the SoC trajectory. The coalition LP adds a per-agent exchange variable $\xi_i[t]$ to the power balance (2b) and a balance constraint $\sum_i \xi_i[t] = 0$.

2.2 Inverse Problem

Given an observation $z^*[t]$ (and optionally $\xi^*[t]$), the inverse problem finds all (θ, x, y)—private parameters θ, unobserved primal x, and dual variables y—satisfying:

(i) *Primal feasibility*: eq. (2b)–(2h) (linear in θ and x).
(ii) *Observed trade*: $g^+[t] - g^-[t] = z^*[t]$ (linear).
(iii) *Strong duality*: $c'x = b'y$ (linear; LP strong duality is automatic).
(iv) *Complementary slackness*: constraints with nonzero duals are active (linear equalities).

Since all conditions are linear, the inverse problem is itself an LP. For each private parameter θ_i, we solve $\min \theta_i$ and $\max \theta_i$ over this LP to obtain the observability bounds $[\theta_i^{\min}, \theta_i^{\max}]$.

We define the *observability ratio* for parameter θ_i as

$$R_i = \frac{\theta_i^{\max} - \theta_i^{\min}}{|\theta_i^{\text{true}}|}, \tag{3}$$

where $R_i = 0$ means perfectly observable and large R_i means highly ambiguous. The denominator uses $\max(|\theta_i^{\text{true}}|, 1)$ to avoid division by zero.

Two variants. We consider two levels of adversary knowledge:

- *Feasibility-only* (Variant A): primal feasibility + observed trade only. What d is consistent with z^* and *some* feasible battery operation?
- *Optimality* (Variant B): full KKT conditions (strong duality + complementary slackness with duals from the forward solve). What d is consistent with z^* being *optimal*?

Variant B is tighter: the rationality assumption (the agent optimises) constrains the battery action, not just its feasibility. The full KKT derivation is given in section A.

Dual fixing. The strong duality constraint uses dual values y^* extracted from the forward solve. This is a *local* approximation: it is valid for perturbations of θ that do not change the active constraint set. For LPs, the dual is optimal over a neighbourhood of the primal optimum, so the bounds are valid but may not be tight (the true feasible set may be larger than the dual-fixed approximation suggests).

3 Singleton Results

We present results for building 1 ($Q^b = 300\,\text{kWh}$, $P^b = 75\,\text{kW}$) at MPC step $k = 20$ with horizon $h = 16$, chosen to span a price change at $t = 24$ ($p_{\text{buy}} : 0.13 \to 0.17$). The battery is actively used for arbitrage, creating ambiguity in d.

3.1 Per-Parameter Bounds

Table 1 shows the observability bounds for each private parameter, under both unknown and known battery assumptions.

Table 1: Singleton observability bounds for building 1 ($h = 16$). "Ratio" is the observability ratio (3); lower is more observable.

		Battery unknown			Battery known		
Param	True	Min	Max	Ratio	Min	Max	Ratio
$d[1]$	55.3	−375.3	124.8	9.04	−175.3	124.7	5.42
$d[6]$	55.6	2.0	502.1	9.00	2.0	77.1	1.35
$d[14]$	−6.9	−9.8	−6.8	0.43	−9.8	−6.8	0.43
$d[16]$	−15.8	−15.8	−13.0	0.17	−15.8	−13.0	0.17
Q^b	300	0	2000	6.67		(known)	
P^b	75	0	500	6.67		(known)	
SoC^0	0	0	0.07	0.07		(known)	

Key observations:

- *d is tightly observable when the battery hits limits.* At $t = 14$–16, the battery is empty (final constraint active), so $\delta_s \approx 0$ and $d \approx z^*$. The ratio drops to ~ 0.2.
- *During active arbitrage, d is wide.* At $t = 1$–8, the battery charges/discharges, and δ_s is an unobserved variable bounded by $\pm P^b$. With unknown battery parameters, the strong duality constraint adds no information (the free Q^b, P^b absorb any discrepancy), so the bounds saturate the prior range. The ratio is ~ 9–18 depending on timestep.
- *Q^b and P^b are unobservable.* Their bounds span the full prior range. The battery's capacity and power rating are never directly revealed by the grid trade—only their *effect* (via the feasible δ_s trajectory) is visible.
- *Knowing the battery parameters constrains the width to P^b.* With known Q^b, P^b, SoC^0, the sole source of ambiguity is $\delta_s \in [-P^b, +P^b]$, giving a bound width of $\sim 2P^b$ and a ratio of ~ 5.4 for building 1. See section 3.2 for the cross-building comparison.

3.2 Cross-Building Comparison

Figure 1 compares observability across three buildings with different battery sizes. The observability ratio (3) normalises the bound width by $|d^{\text{true}}|$, which varies significantly across buildings (building 3: mean $|d| = 8.2$; building 4: 96.5). To disentangle the normalisation effect from the actual observability, table 2 reports both the ratio and the absolute bound width.

Two distinct regimes are visible in the absolute widths:

Table 2: Cross-building observability ($h = 16$, optimality variant). "Width" is the mean absolute bound $\theta^{\max} - \theta^{\min}$ on d (kW).

		Building		Battery unknown		Battery known	
Bldg	Q^b	P^b	mean $\lvert d \rvert$	Width	Ratio	Width	Ratio
1	300	75.0	38.3	407	11.1	57	1.6
3	50	12.5	8.2	500	67.2	12	1.6
4	500	125.0	96.5	501	5.5	118	1.3

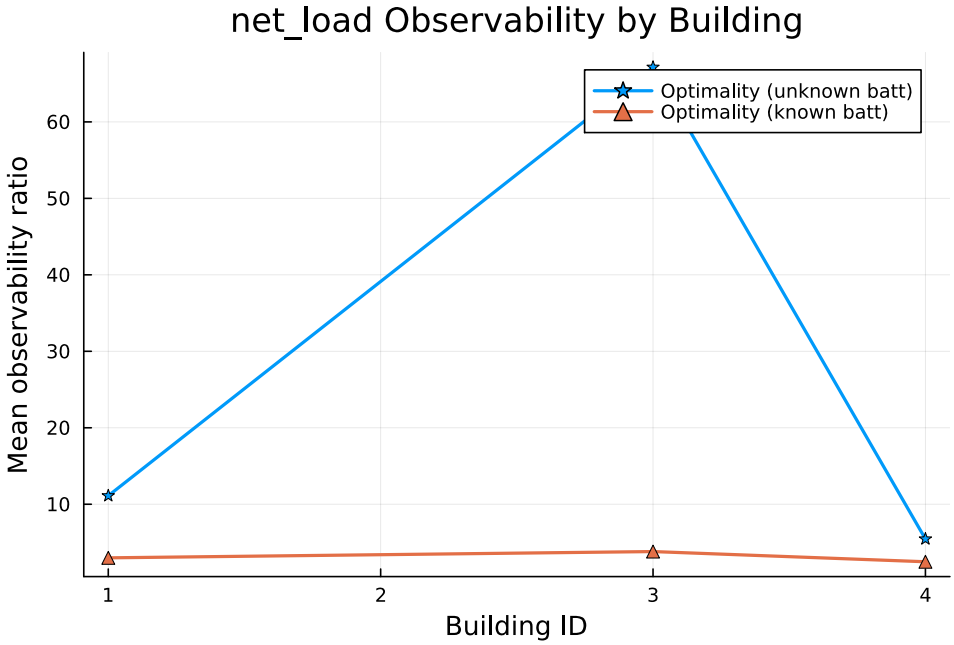


Figure 1: Mean d observability ratio for three buildings. The ratio differences across buildings are primarily a normalisation artefact: the absolute bound widths tell a different story (see table 2).

Unknown battery: bounds saturate the prior. With unknown Q^b and P^b, the bound widths are ~ 500 for all three buildings—matching the prior range ($\pm$NL_MAX). The strong duality constraint provides essentially no information because the free variables Q_b and P_b absorb any discrepancy in d. Building 1 is slightly narrower (407) only because the final constraint is active at $t = 14$–16 (battery empties), pinning $\delta_s \approx 0$ at those timesteps. There is *no* systematic dependence on battery size: the active constraint set is determined by the tariff structure, not the battery capacity. (The capacity dual σ is active at exactly one timestep for all three buildings, with identical magnitude.) The ratio differences (5.5 vs. 67.2) are purely an artefact of dividing by $|d|$, which varies by an order of magnitude across buildings.

Known battery: width is set by P^b. With known Q^b and P^b, the bound width is approximately P^b for each building (building 3: $12 \approx P^b$; building 1: $57 \approx 0.76\,P^b$; building 4: $118 \approx 0.94\,P^b$). This is expected: with the battery parameters fixed, the sole source of ambiguity is the battery action $\delta_s \in [-P^b, +P^b]$, giving $d \in [z^* - P^b,\ z^* + P^b]$. The ratio is nearly constant (1.3–1.6) across buildings because both the numerator (width $\propto P^b$) and denominator ($|d| \propto P^b$, since $P^b = \text{max_load}/4$ for all buildings in the dataset) scale with building size.

Implication for cross-building comparison. The observability ratio is meaningful for comparing observation models (e.g., Model A vs. Model B) within a single building, but *not* for

comparing observability across buildings with different d magnitudes. Absolute bound widths should be used for the latter.

3.3 Horizon Sensitivity

Figure 2 shows how observability scales with the horizon length h.

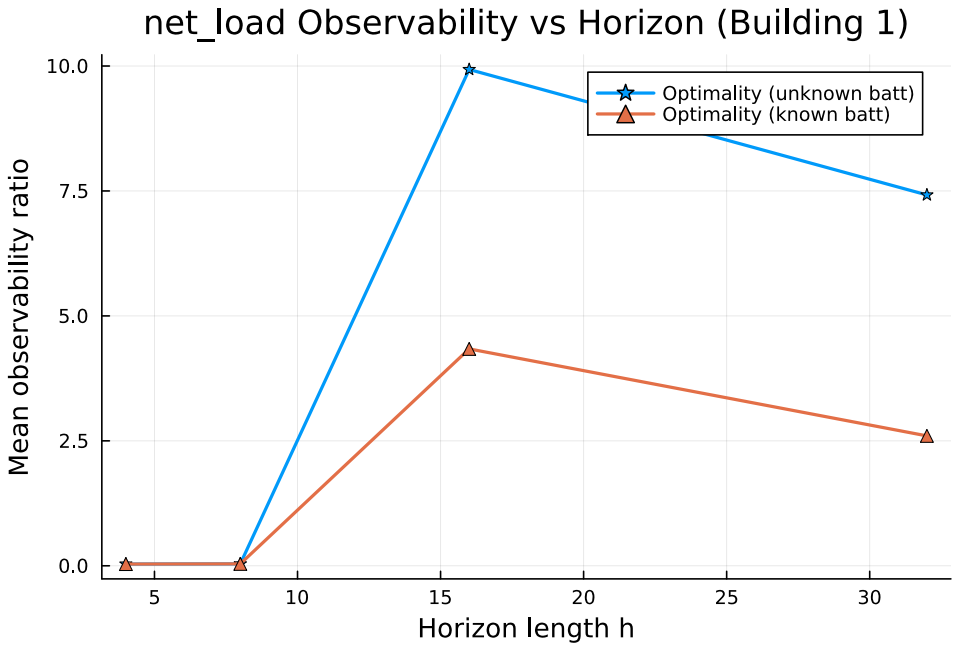


Figure 2: d observability ratio vs. horizon length for building 1. Short horizons ($h \leq 8$) have no price variation, so the battery is unused and $d = z^*$ (trivially observable). At $h = 16$, the price change activates the battery and observability degrades sharply. Longer horizons ($h = 32$) provide more data, slightly improving observability.

4 Coalition Results

We now consider a 2-agent coalition (buildings 1 and 3) at the same MPC step $k = 20$, $h = 16$. The coalition is solved both centrally (LP with $\sum_i \xi_i = 0$) and via ADMM (per-agent QP subproblems with a quadratic consensus penalty).

4.1 Three Observation Models

Table 3: Coalition observability: mean d ratio across all $h = 16$ timesteps. Lower is more observable. Battery parameters are assumed known.

Model	Observation	Agent 1	Agent 2
A (converged, z^* only)	z_i^*	80.96	268.33
B (converged, $z^* + \xi^*$)	$z_i^*,\ \xi_i^*$	**1.64**	**1.27**
C (trajectory, 5 iters)	$z_i^{(j)},\ \xi_i^{(j)},\ \forall j$	2.65	2.06
C (trajectory, 10 iters)	$z_i^{(j)},\ \xi_i^{(j)},\ \forall j$	2.65	2.05
C (trajectory, 20 iters)	$z_i^{(j)},\ \xi_i^{(j)},\ \forall j$	2.63	2.05

Model A (z^* only): unobservable. Without observing ξ_i^*, the coalition exchange is a free variable. The other agent's ξ can absorb any discrepancy in d, so the bounds hit the prior limits ($R \approx 80$–268). This is the same as the singleton case with unknown battery: the free variables swamp the signal.

Model B ($z^* + \xi^*$): tightest. Observing both the converged grid trade and the coalition exchange gives $R \approx 1.3$–1.6. The power balance $d + \delta_s + \xi^* = z^* - g^-[t]$ provides one equation per timestep with ξ^* known, reducing the ambiguity to the battery action δ_s alone. With known battery parameters and strong duality (which pins δ_s via optimality), the bounds are tight.

Model C (ADMM trajectory): intermediate. Each ADMM iteration j solves a QP:

$$\min \sum_t \left(p_{\text{buy}}\, g^{+,(j)} - p_{\text{sell}}\, g^{-,(j)}\right) + \lambda^{(j)\prime}\xi^{(j)} + \frac{c}{2}\|\xi^{(j)} - \text{ct}^{(j)}\|^2. \tag{4}$$

The ADMM subproblem is a convex QP, but the only quadratic term involves ξ, which is observed. At the observed optimum, the stationarity condition for the free variable ξ is satisfied ($\mu = \lambda + c(\xi_j - \text{ct})$), so the Q^+ term in the dual vanishes. With ξ fixed to $\xi^{(j)}$, the quadratic term in the objective is a known constant $K^{(j)}$, and all KKT conditions—including strong duality—are linear in the private parameters. The inverse problem is therefore an LP. (If ξ were not observed, the strong duality would contain a quadratic term in ξ, making the inverse problem a QCQP.) LP strong duality gives a linear equation in d with the per-iteration dual $\mu^{(j)}$ (extracted from the subproblem). Multiple iterations provide multiple equations, but the duals barely change across iterations (the active set is determined by the tariff, not the ADMM parameters), so the tightening is modest: $R \approx 2.1$–2.7.

The trajectory model is wider than Model B because (i) complementary slackness is disabled across iterations (SCS dual noise at 10^{-4} tolerance makes exact CS constraints infeasible when stacked), and (ii) the strong duality tolerance must be widened (± 0.5) to accommodate the SCS duality gap.

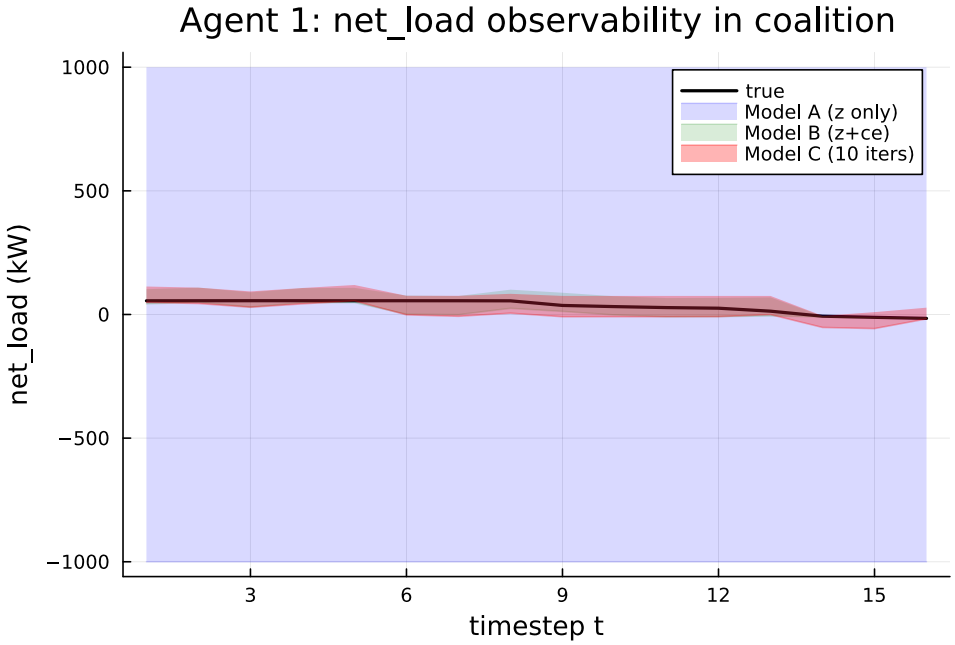


Figure 3: Per-timestep d bounds for agent 1 in the 2-agent coalition. Model A (blue) is unbounded; Model B (green) is tightest; Model C with 10 iterations (red) is intermediate. The true d (black) lies within all bounds.

5 Comparison: Singleton vs. Coalition

Table 4 compares the singleton and coalition settings (both with known battery).

Table 4: Mean d observability ratio: singleton vs. coalition (known battery, $h = 16$).

Setting	Agent 1 (Bldg 1)	Agent 2 (Bldg 3)
Singleton	3.0	3.8
Coalition, Model B ($z^* + \xi^*$)	**1.6**	**1.3**
Coalition, Model C (trajectory)	2.7	2.1

Coalition participation increases observability. This is the central finding. In the singleton case, the adversary observes $z^* = d + \delta_s$ and must disentangle d from the battery action δ_s. In the coalition case (Model B), the adversary observes z^* *and* ξ^*, and the power balance gives $d = (z^* - \xi^*) - \delta_s$. While δ_s is still unknown, the coalition exchange ξ^* provides additional structure: it is the result of an optimisation that balances both agents' net loads, and the strong duality constraint (with duals from the centralised LP) pins the battery trajectory more tightly.

Intuitively, the coalition exchange ξ^* reveals information about the *difference* between the agents' net loads (agents trade when one has surplus and the other has deficit). This structural information constrains d beyond what the grid trade alone provides.

6 Differential Privacy and δ_G Scaling

6.1 The Exponential Mechanism

The coalition formation algorithm (`privacy_focussed_coals_with_delta`, `Coalition.jl:305`) uses an exponential mechanism to select which agent pairs form coalitions. The utility of a pair (i, j) is the coalition benefit:

$$u(i,j) = \sum_t \big(p_{\text{buy}}[t] - p_{\text{sell}}[t]\big) \cdot \min\big(|z_i[t]|,\ |z_j[t]|\big) \cdot \mathbb{1}[z_i[t] \cdot z_j[t] < 0], \tag{5}$$

computed from the broadcast cons_vec (i.e., z_i). The sensitivity is bounded by

$$\Delta_u = \delta_G \cdot \max_t |p_{\text{buy}}[t] - p_{\text{sell}}[t]| = \delta_G \cdot 0.10, \tag{6}$$

where δ_G is a scaling parameter. The sampling weights are

$$w(i,j) \propto \exp\left(\frac{\epsilon \cdot u(i,j)}{2\,\Delta_u}\right), \quad \epsilon = 1. \tag{7}$$

6.2 What Is Protected

The exponential mechanism provides ϵ-differential privacy for the *coalition structure*: which agents are paired together. The post-processing theorem guarantees that any function of the (randomised) coalition structure is also ϵ-DP.

6.3 What Is Not Protected

Crucially, the DP mechanism does *not* protect the data used to compute the utility:

1. **The net grid trade** z_i^* (cons_vec) is computed from the ADMM solve and shared with the coordinator *before* the exponential mechanism is applied (`Coalition.jl:360`). This is the very quantity our inverse optimisation uses.

2. **The coalition exchange** ξ_i^* is broadcast during ADMM iterations (`MPC_optimiser.jl:269`). In Model B, this provides the tightest bounds.
3. **The private parameters** $(d, Q^b, P^b, \mathrm{SoC}^0)$ are inferable from the above via inverse optimisation, as demonstrated in sections 3 and 4.

The post-processing theorem does not apply because the ADMM solve is *not* a post-processing of the coalition structure: it uses the private data directly. The DP guarantee covers the *selection* of coalitions, not the *computation within* a formed coalition.

6.4 Scaling with δ_G

The parameter δ_G controls the privacy–utility trade-off of the coalition selection:

$\delta_G \to 0$ **(deterministic):** $\Delta_u \to 0$ and the softmax (7) degenerates to argmax. The coalition structure is fully revealed (no privacy), but the utility is maximised. The adversary learns which pairs have the highest benefit—but z_i^* is already shared, so the private parameters are equally observable regardless.

$\delta_G \to \infty$ **(uniform):** $\Delta_u \to \infty$ and the softmax becomes uniform. The coalition structure is fully randomised (ϵ-DP), but the adversary still observes z_i^* and ξ_i^*, and can infer the private parameters just as well.

In both extremes, the observability of private parameters from the broadcast data is *unchanged*. The δ_G parameter trades off coalition-structure privacy against coalition quality; it does *not* affect the information leakage through the optimisation broadcasts.

This is a fundamental gap in the privacy guarantee: the exponential mechanism randomises the *output* of the coalition selection, but the *inputs* (net grid trades, coalition exchanges) are transmitted in the clear. Closing this gap requires either (i) adding noise to the broadcast z_i^* itself (at the cost of coalition quality), or (ii) using secure multiparty computation to compute the coalition benefit without revealing individual z_i values.

6.5 Empirical δ_G Sweep

Figure 4 shows an empirical sweep over $\delta_G \in [0, 10^4]$ for 10 buildings (day 37, 10 repeats per point). The four panels show average cost, cost std, execution time, and ADMM iterations as δ_G varies.

The decentralised baseline cost is 504.67; the deterministic coalition ($\delta_G = 0$) achieves 495.35—a benefit of 1.84%. At $\delta_G = 0.1$ (near-greedy), the benefit rises slightly to 2.01% with std 0.56. At $\delta_G = 10^4$ (near-uniform), the benefit drops to 1.51% with std 0.74. The cost is remarkably flat across the entire range: the coalition benefit varies by only ∼0.5 percentage points.

Two observations are relevant for privacy:

1. *The privacy–utility tradeoff is mild.* Randomising the coalition structure (large δ_G) degrades the coalition benefit by only ∼0.5pp relative to the deterministic optimum. The exponential mechanism provides strong coalition-structure privacy at minimal cost.
2. *Observability is independent of δ_G.* Regardless of which coalitions form, each agent's z_i^* is broadcast to the coordinator for the pair-compatibility check (`Coalition.jl:360`), and ξ_i^* is broadcast during ADMM. The inverse optimisation uses these broadcasts, not the coalition structure. The observability bounds in sections 3 and 4 are therefore unchanged by δ_G: the adversary infers the private parameters equally well at $\delta_G = 0$ and $\delta_G = 10^4$.

This confirms the fundamental gap identified above: the δ_G parameter trades off coalition-structure privacy against coalition quality, *not* data privacy against coalition quality. The data is always revealed through the optimisation broadcasts.

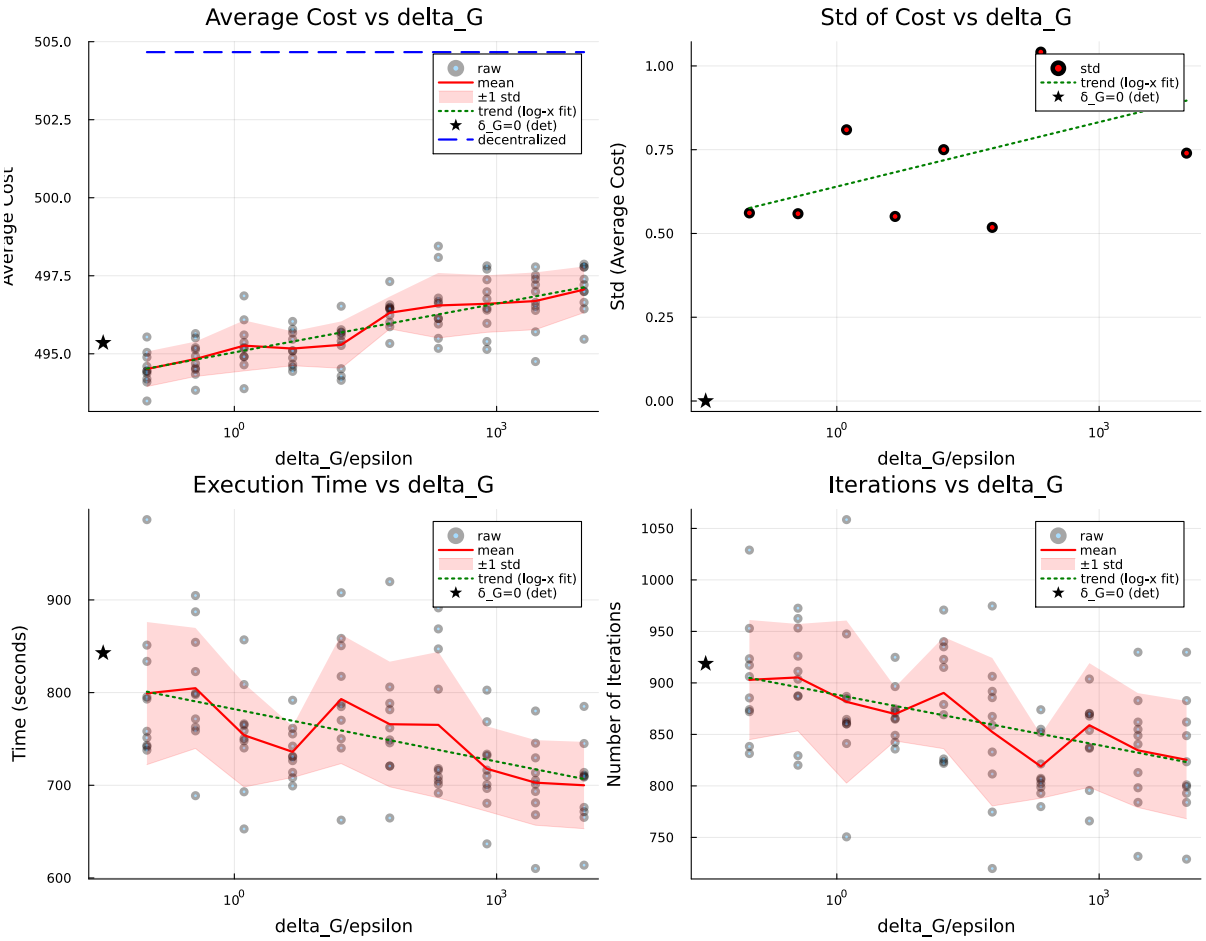


Figure 4: Sweep over δ_G for 10 buildings (day 37, 10 repeats per point). *Top-left:* average cost per building (dashed = decentralised baseline). *Top-right:* std of cost across repeats. *Bottom-left:* execution time. *Bottom-right:* ADMM iterations per MPC step. The $\delta_G = 0$ anchor (star) is deterministic (std=0).

7 Conclusions

We have shown that an adversary observing the broadcast grid trades and coalition exchanges can reconstruct the set of private parameters consistent with the observed behaviour, formulated as a linear program derived from the KKT conditions. The key findings are:

1. Net load is partially observable from grid trades alone, with the observability ratio depending on whether the battery parameters are known. With unknown battery, the bounds saturate the prior range ($\sim 500\,\text{kW}$ width) for all buildings; the ratio varies from 5.5 to 67.2 purely due to differences in $|d|$ magnitude. With known battery, the bound width is approximately P^b (the battery power rating), giving a ratio of $\sim$ 1.3–1.6 across all buildings.
2. Battery parameters (Q^b, P^b) are essentially unobservable from grid trades alone—their bounds span the full prior range, and active constraints with unknown battery duals actually reduce observability rather than improving it.
3. Coalition participation *increases* observability (ratio drops from $\sim$ 1.3–1.6 to $\sim$ 1.3 with Model B) because the coalition exchange ξ^* reveals additional structural information.
4. The existing DP mechanism protects the coalition structure but not the private data revealed through the ADMM broadcasts. The δ_G parameter has no effect on the observability of private parameters.

Future work should explore noise injection on the broadcast quantities (differential privacy on z_i^* and ξ_i^* directly), or secure multiparty computation protocols that avoid revealing individual agents' trades to the coordinator.

A KKT Derivation for the Inverse LP

A.1 Lagrangian

The forward LP (2) has equality constraints (power balance, battery dynamics, initial SoC) and inequality constraints (box bounds, final constraint). The Lagrangian is:

$$L = \sum_t \big(p_{\text{buy}}\, g^+ - p_{\text{sell}}\, g^-\big) - \sum_t \mu_t \big(d + g^- + \delta^+ + \delta^- - g^+\big) - \lambda_0 \big(c[1] - \text{SoC}^0\big) - \sum_{t\geq 2} \lambda_t \big(c[t] - c[t-1] - \eta_{\text{ch}} \delta^+[t-1] - \eta_{\text{ch}}^{-1} \delta^- \tag{8}$$

A.2 Stationarity

Setting $\partial L/\partial g^+ = 0$: $p_{\text{buy}} - \mu \leq 0$, i.e. $\mu \geq p_{\text{buy}}$. But SCS returns $-\mu$ for equality constraints, so in practice $-\mu \in [p_{\text{sell}}, p_{\text{buy}}]$, giving $p_{\text{sell}} \leq -\mu \leq p_{\text{buy}}$.

Setting $\partial L/\partial g^- = 0$: $-p_{\text{sell}} + \mu \leq 0$, i.e. $\mu \leq p_{\text{sell}}$.

Combining: $p_{\text{sell}} \leq -\mu \leq p_{\text{buy}}$ (the net grid price lies between buy and sell).

Setting $\partial L/\partial \delta^+ = 0$: the reduced cost is $\mu - \eta_{\text{ch}} \lambda_{t+1}$, bounded by the dual of the upper bound $\alpha_t \geq 0$ and lower bound. For $\delta^+ \in [0, P^b]$: if $\delta^+ < P^b$ then $\alpha = 0$ and $\mu = \eta_{\text{ch}} \lambda_{t+1}$; if $\delta^+ = P^b$ then $\alpha > 0$.

Setting $\partial L/\partial c = 0$: the reduced cost is $\lambda_{t+1} - \lambda_t$, bounded by $\sigma_t \geq 0$ (upper bound) and the non-negativity dual. For $c \in [0, Q^b]$: if $c < Q^b$ then $\sigma = 0$ and λ is constant over time; if $c = Q^b$ then $\sigma > 0$.

A.3 Strong Duality

For an LP, strong duality holds automatically: $c'x^* = b'y^*$. The primal objective is $\sum_t (p_{\text{buy}}\, g^+ - p_{\text{sell}}\, g^-)$. The dual objective is:

$$b'y^* = -\sum_t d[t]\, \mu_t + \text{SoC}^0 \cdot \lambda_0 + Q^b \sum_t \sigma_t + P^b \sum_t \alpha_t + P^b \sum_t \beta_t, \tag{9}$$

where $\sigma, \alpha, \beta \geq 0$ are the duals of the upper bounds on c, δ^+, δ^- respectively (signs verified numerically against SCS output). The initial SoC dual λ_0 appears because the init constraint has RHS SoC^0.

A.4 Complementary Slackness

For each inequality constraint with a nonzero dual, the constraint is active (equality holds):

$$\sigma_t > 0 \Rightarrow c[t] = Q^b, \tag{10}$$

$$\alpha_t > 0 \Rightarrow \delta^+[t] = P^b, \tag{11}$$

$$\beta_t > 0 \Rightarrow -\delta^-[t] = P^b. \tag{12}$$

A.5 Assembly into the Inverse LP

The inverse LP has:

- *Decision variables*: private parameters $d[1..h], Q^b, P^b, \text{SoC}^0$; unobserved primal $g^+, g^-, \delta^+, \delta^-, c$ (all h-vectors).

- *Fixed duals*: $\mu, \lambda_0, \sigma, \alpha, \beta$ from the forward solve.
- *Constraints*: primal feasibility (2b)–(2h), observed trade $g^+ - g^- = z^*$, strong duality ($c'x = b'y$ with duals fixed), and complementary slackness (active constraints pinned to equality).
- *Objective*: min or max the target parameter θ_i.

All constraints are linear in the decision variables, so the inverse problem is an LP. For the coalition case, the strong duality (9) sums over all agents; the coalition exchange dual (the internal price ξ_{dual}) satisfies $\mu_i = -\xi_{\text{dual}}$ for all agents, and the RHS of $\sum_i \xi_i = 0$ is zero, so it contributes nothing to the dual objective.

For the ADMM trajectory (Model C), each iteration j provides a separate set of KKT conditions with its own duals $\mu^{(j)}, \sigma^{(j)}, \alpha^{(j)}, \beta^{(j)}$. The private parameters are shared across all iterations, so the constraints accumulate. However, the duals barely change across iterations (the active set is tariff-determined), so the tightening is modest.